

Customer Behavior Analysis

A comprehensive exploration of shopping patterns, revenue insights, and customer segmentation using SQL queries and data visualization techniques.

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Dataset Foundation

Our analysis examines 3,900 customer records across 18 key attributes including demographics, purchase behavior, and preferences. The dataset captures comprehensive shopping patterns from customer ID and age to payment methods and purchase frequency.

After data cleaning, we addressed 37 missing review ratings by filling them with category-specific median values. We also engineered new features including age groups (Young Adult, Adult, Middle Aged, Senior) and purchase frequency in days to enable deeper analysis.

3.9K

Total Customers

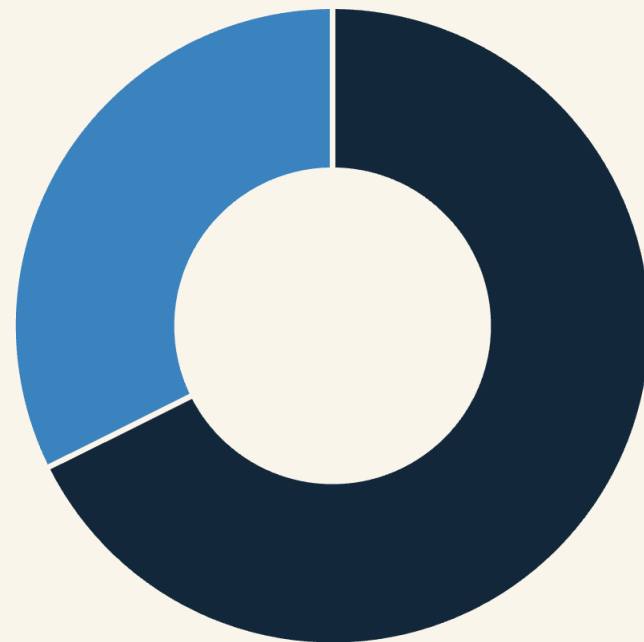
18

Data Attributes

50

IND Locations

Gender Revenue Comparison



■ Male ■ Female

Male Customers Drive Revenue

Male customers generated 157,890 in total revenue compared to 75,191 from female customers, representing a 2.1x difference. This significant gap suggests male customers either make more frequent purchases or spend more per transaction.

The gender distribution shows males comprise 68% of the customer base (2,652 customers) versus 32% female (1,248 customers), which partially explains the revenue disparity.



High-Value Discount Shoppers

Analysis reveals 9 customers who used discounts yet still spent above the average purchase amount of \$59.76. These savvy shoppers demonstrate that discount usage doesn't necessarily correlate with lower spending.

Customer IDs

2, 3, 4, 7, 9, 12, 13, 16

Purchase Range

64 to 97 per transaction

Key Insight

Discounts attract high spenders too

Shipping Type Performance

Comparing average purchase amounts between Standard and Express shipping reveals minimal difference. Express shipping customers average \$60.48 per purchase, while Standard shipping customers average \$58.46—a difference of just \$2.02.

This suggests shipping preference is driven by convenience rather than purchase value, and both shipping types attract similar customer segments in terms of spending behavior.

Express Shipping

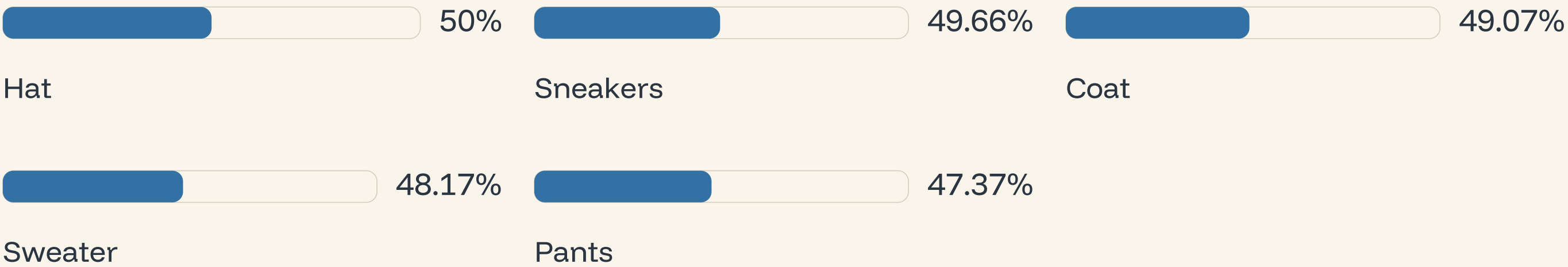
Average: 60.48

Standard Shipping

Average: 58.46

Top 5 Products with Highest Discount Rates

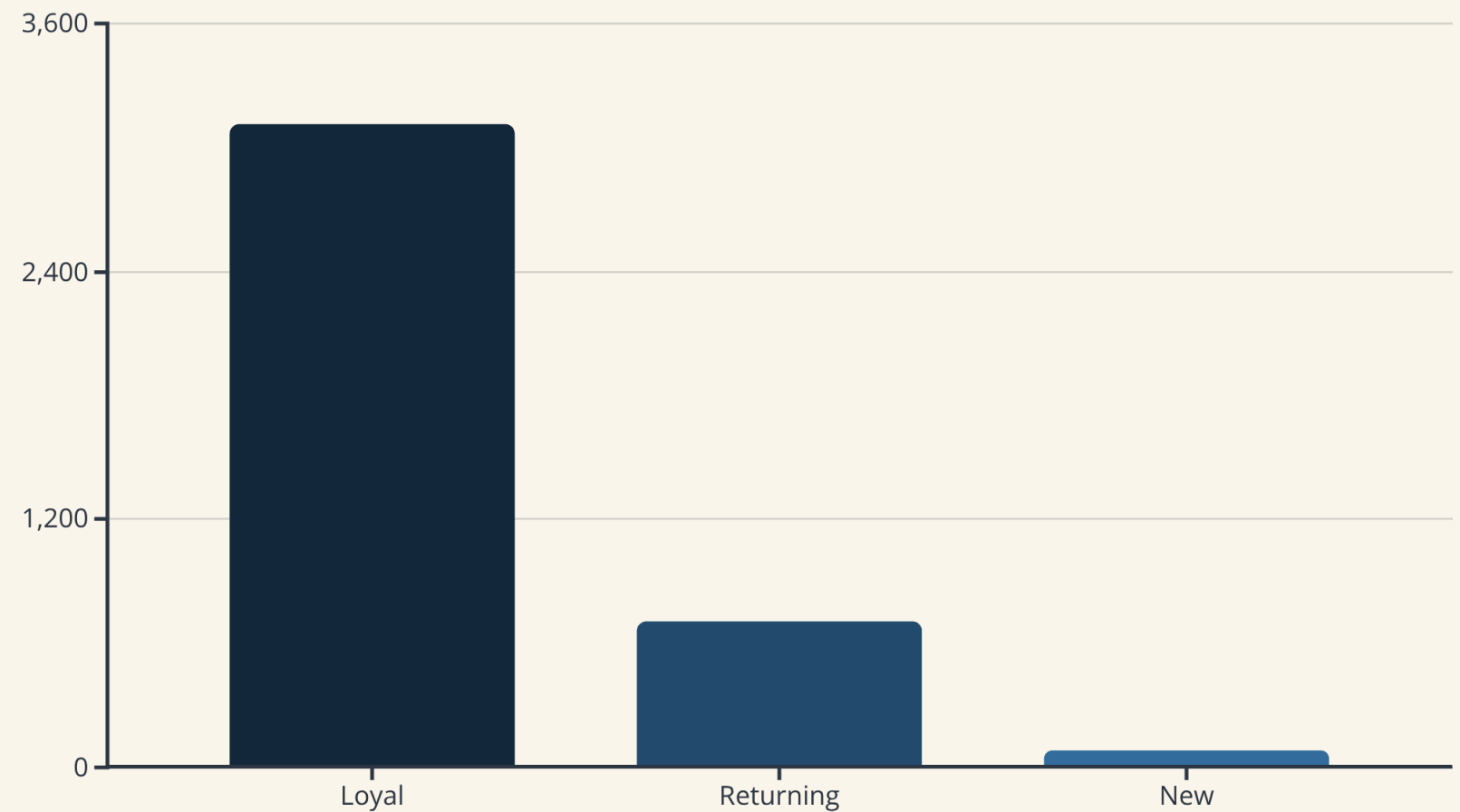
Analysis of discount application rates across products reveals which items are most frequently purchased with discounts. Hats lead with 50% of purchases using discounts, followed closely by Sneakers at 49.66%.



Customer Loyalty Segments

Customers are segmented based on previous purchase history: New (1 purchase), Returning (2-10 purchases), and Loyal (11+ purchases).

The data reveals a highly loyal customer base, with 79.9% classified as Loyal customers, demonstrating strong retention and repeat purchase behavior.



Top 3 Products by Category

Using window functions, we identified the most popular items within each product category. This analysis reveals customer preferences and helps optimize inventory management.



Accessories

1. Jewelry (171 orders)
2. Sunglasses (161)
3. Belt (161)



Clothing

1. Blouse (171 orders)
2. Pants (171)
3. Shirt (169)



Footwear

1. Sandals (160 orders)
2. Shoes (150)
3. Sneakers (145)

Subscription Status Impact

The Subscription Paradox

Contrary to expectations, non-subscribers actually spend slightly more on average (59.87) compared to subscribers (59.49). However, the total revenue tells a different story.

Non-subscribers generate 170,436 in total revenue with 2,847 customers, while subscribers contribute 62,645 with 1,053 customers. The higher non-subscriber revenue is driven by volume rather than individual spending.

2847

Non-Subscribers

59.87 avg spend

1053

Subscribers

59.49 avg spend

Strategic Insights & Recommendations



Gender-Focused Marketing

Male customers drive 68% of revenue. Develop targeted campaigns while exploring opportunities to increase female customer engagement and spending.



Leverage Loyalty

With 79.9% loyal customers, focus on retention programs and exclusive benefits to maintain this strong base while converting returning customers to loyal status.



Smart Discounting

Discounts attract high-value customers too. Strategic discount campaigns on top products like Hats and Sneakers can drive volume without sacrificing margins.



Category Optimization

Clothing dominates with 104K revenue. Expand popular items like Blouses and Pants while testing new products in high-performing categories.